5.2 The Roots-of-Unity Filter

WORKSHEET

The question

STEP I 1990, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q2

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself ω is a root of $z^3 - 1 = 0$. Factorise $z^3 - 1$. Why does this give $1 + \omega + \omega^2 = 0$?

STAGE 2 The filter

Ask yourself What is $1 + \omega^k + \omega^{2k}$ when k is a multiple of 3? When it is not?

Pause and try Add up $(1 + \omega^r)^n$ for $r = 1, 2, 3$, expanding each by the binomial theorem. What do you get?

STAGE 3 Two calculations

Ask yourself For the second sum we need positions $k \equiv 2 \pmod 3$. How can you modify the filter to detect $k \equiv 2$ instead of $k \equiv 0$?

What's really going on

The discrete Fourier transform

Replace 3 by any N and ω by $e^{2\pi i/N}$. For a list of numbers $a_0, a_1, \ldots, a_{N-1}$, the numbers

$$\hat{a}_r = \sum_{k=0}^{N-1} a_k \, \omega^{rk}$$

form its *discrete Fourier transform*, and the filter identity $\sum_r \omega^{rk} = N$ or 0 is what lets you recover the a_k from the $\hat{a}_r$. The transform splits a signal into its component frequencies. In 1965 James Cooley and John Tukey published a way to compute it in about $N \log_2 N$ steps instead of N^2, the *fast Fourier transform*; Gauss had found essentially the same idea in 1805 but never published it. The FFT is used in audio and image compression (MP3 and JPEG), in mobile phone signals, in medical scanners and in multiplying enormous numbers, and it is often listed among the most important algorithms of the twentieth century.

Generating functions

The filter is one instance of a bigger idea: a sequence $a_0, a_1, a_2, \ldots$ can be packaged as a polynomial or power series $A(x) = \sum a_k x^k$, its *generating function*, and operations on the sequence become operations on the function. Evaluating at 1 sums the sequence; evaluating at roots of unity picks out residue classes; differentiating weights the terms by k. This turns many counting problems into algebra, and it is a central technique in Chapter 9. The same filter, applied to $(1+x)^n$ with fifth roots of unity, connects to the regular pentagon and the golden ratio; we pursue the algebra of roots of unity in Section 6.1.

Try this next

1 **Even positions.** Use the filter with square roots of unity (1 and -1) to show that $\binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \cdots = 2^{n-1}$.

2 **The other residue.** Find a formula for $\binom{n}{1} + \binom{n}{4} + \binom{n}{7} + \cdots$, and check that the three formulas add up to 2^n.

3 **Every fourth.** Using i, show that $\binom{n}{0} + \binom{n}{4} + \binom{n}{8} + \cdots = \frac{1}{4}\left(2^n + 2^{n/2+1}\cos\frac{n\pi}{4}\right)$.

4 **Divisibility.** How many subsets of $\{1, 2, \ldots, 12\}$ have a number of elements divisible by 3? How many have a sum divisible by 3? (*Hint for the second:* consider $\prod_{j=1}^{12}(1 + x^j)$ at the cube roots of unity.)

Further reading

- **Herbert Wilf, *generatingfunctionology*** (A. K. Peters; free online). A delightful introduction to generating functions; the roots-of-unity filter appears early on.
- **Arthur Benjamin & Jennifer Quinn, *Proofs That Really Count*** (Mathematical Association of America). Binomial identities proved by counting.
- **T. W. Körner, *Fourier Analysis*** (Cambridge University Press). Includes the discrete Fourier transform and the FFT.

The full guided solution is in Section 5.2:
https://beyondstep.keithmcnulty.org/ch05/roots-filter.html